

Advanced Policy Gradient

CS 185/285

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Part 1:

Which shortcuts did we take so far?

The off-policy policy gradient

$$\theta^* = \arg \max_{\theta} J(\theta) \quad J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[r(\tau)]$$

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \nabla_{\theta'} \log \pi_{\theta'}(\tau) r(\tau) \right]$$

$$= E_{\tau \sim p_{\theta}(\tau)} \left[\left(\prod_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \right) \left(\sum_{t=1}^H \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \right) \left(\sum_{t=1}^H r(\mathbf{s}_t, \mathbf{a}_t) \right) \right]$$

$$= E_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^H \left(\prod_{t'=1}^t \frac{\pi_{\theta'}(\mathbf{a}_{t'} | \mathbf{s}_{t'})}{\pi_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})} \right) \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \left(\sum_{t'=t}^H r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \left(\prod_{t''=t}^{t'} \frac{\pi_{\theta'}(\mathbf{a}_{t''} | \mathbf{s}_{t''})}{\pi_{\theta}(\mathbf{a}_{t''} | \mathbf{s}_{t''})} \right) \right) \right]$$

$$= E_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^H \underbrace{\left(\frac{\cancel{\pi_{\theta'}(\mathbf{s}_t)}}{\cancel{\pi_{\theta}(\mathbf{s}_t)}} \frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \right)}_{\text{use } \pi_{\theta}(\mathbf{s}_t) \text{ instead of } \pi_{\theta'}(\mathbf{s}_t)} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \left(\sum_{t'=t}^H r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \underbrace{\left(\prod_{t''=t}^{t'} \frac{\cancel{\pi_{\theta'}(\mathbf{a}_{t''} | \mathbf{s}_{t''})}}{\cancel{\pi_{\theta}(\mathbf{a}_{t''} | \mathbf{s}_{t''})}} \right)}_{\text{use } \hat{A}^{\pi_{\theta}} \text{ instead of } \hat{A}^{\pi_{\theta'}}} \right) \right]$$

Why is this (kind of) OK to do?

use $\pi_{\theta}(\mathbf{s}_t)$ instead of $\pi_{\theta'}(\mathbf{s}_t)$

use $\hat{A}^{\pi_{\theta}}$ instead of $\hat{A}^{\pi_{\theta'}}$

Today

- Discuss why it's OK to use the wrong advantage
- Discuss why it's OK to use the wrong state distribution
- What kind of algorithms does this motivate
- How are the algorithms we use related to this
- What other algorithms can we derive
- What does this tell us about how we should use policy gradients?

Part 2:

From policy iteration to policy gradient

Why does policy gradient work?

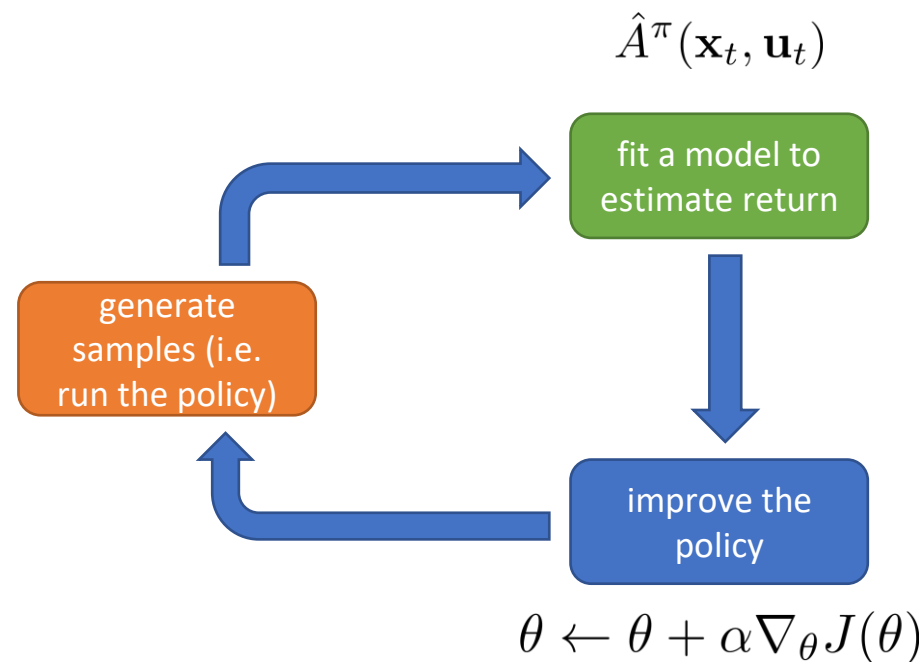
$$\nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t}) \hat{A}_{i,t}^{\pi}$$

- 
1. Estimate $\hat{A}^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$ for current policy π
 2. Use $\hat{A}^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$ to get *improved* policy π'

look familiar?

policy iteration algorithm:

- 
1. evaluate $A^{\pi}(\mathbf{s}, \mathbf{a})$
 2. set $\pi \leftarrow \pi'$



Policy gradient as **policy iteration**

$$J(\theta) = E_{\tau \sim p_{\theta}(\tau)} \left[\sum_t \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

$$J(\theta') - J(\theta) = J(\theta') - E_{\mathbf{s}_0 \sim p(\mathbf{s}_0)} [V^{\pi_{\theta}}(\mathbf{s}_0)]$$

$$= J(\theta') - E_{\tau \sim p_{\theta'}(\tau)} [V^{\pi_{\theta}}(\mathbf{s}_0)]$$

$$\text{claim: } J(\theta') - J(\theta) = E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right]$$

$$= J(\theta') - E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t V^{\pi_{\theta}}(\mathbf{s}_t) - \sum_{t=1}^{\infty} \gamma^t V^{\pi_{\theta}}(\mathbf{s}_t) \right]$$

$$= J(\theta') + E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right] + E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (r(\mathbf{s}_t, \mathbf{a}_t) + \gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right]$$

Policy gradient as policy iteration

$$J(\theta') - J(\theta) = E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right]$$

expectation under $\pi_{\theta'}$

advantage under π_θ

importance sampling

$$\begin{aligned} E_{x \sim p(x)}[f(x)] &= \int p(x) f(x) dx \\ &= \int \frac{q(x)}{q(x)} p(x) f(x) dx \\ &= \int q(x) \frac{p(x)}{q(x)} f(x) dx \\ &= E_{x \sim q(x)} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

$$E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] = \sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)} \left[\gamma^t A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

$$= \sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \gamma^t A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

but we still need states from $p_{\theta'}(\mathbf{s}_t)$

so it's OK not to use $A^{\pi_{\theta'}}(\mathbf{s}_t, \mathbf{a}_t)$

we justified one of the two “hacks”

but the second one is harder!

Part 3:

When is it OK to ignore the state distribution?

Ignoring distribution mismatch?

$$\sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] \stackrel{?}{\approx} \underbrace{\sum_t E_{\mathbf{s}_t \sim p_{\theta}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]}_{\bar{A}(\theta')}$$

why do we want this to be true?

$$J(\theta') - J(\theta) \approx \bar{A}(\theta') \quad \Rightarrow \quad \theta' \leftarrow \arg \max_{\theta'} \bar{A}(\theta')$$

2. Use $\hat{A}^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$ to get *improved* policy π'

is it true? and when?

Claim: $p_{\theta}(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_{θ} is *close* to $\pi_{\theta'}$

Bounding the distribution change

Claim: $p_\theta(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_θ is *close* to $\pi_{\theta'}$

Simple case: assume π_θ is a *deterministic* policy $\mathbf{a}_t = \pi_\theta(\mathbf{s}_t)$

$\pi_{\theta'}$ is *close* to π_θ if $\pi_{\theta'}(\mathbf{a}_t \neq \pi_\theta(\mathbf{s}_t) | \mathbf{s}_t) \leq \epsilon$

$$p_{\theta'}(\mathbf{s}_t) = \underbrace{(1 - \epsilon)^t}_{\text{probability we made no mistakes}} p_\theta(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) \underbrace{p_{\text{mistake}}(\mathbf{s}_t)}_{\text{some other distribution}}$$

seem familiar?

$$\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| = (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{mistake}}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| \leq 1 - (1 - \epsilon)^t$$

useful identity: $(1 - \epsilon)^t \geq 1 - \epsilon t$ for $\epsilon \in [0, 1]$ $\leq \epsilon t$

not a great bound, but a bound!

Bounding the distribution change

Claim: $p_\theta(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_θ is *close* to $\pi_{\theta'}$

General case: assume π_θ is an arbitrary distribution

$\pi_{\theta'}$ is *close* to π_θ if $\frac{1}{2} \sum_{\mathbf{s}_t} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)| = D_{\text{TV}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t), \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$

Useful lemma: if $D_{\text{TV}}(p_X(x), p_Y(y)) \leq \epsilon$, exists $p(x, y)$ such that $p(x) = p_X(x)$, $p(y) = p_Y(y)$ and $p(x = y) = 1 - \epsilon$

$\Rightarrow p_X(x)$ “agrees” with $p_Y(y)$ with probability ϵ

$\Rightarrow \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)$ takes a different action than $\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ with probability at most ϵ

$$\begin{aligned} \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| &= (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{mistake}}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| \leq 1 - (1 - \epsilon)^t \\ &\leq \epsilon t \end{aligned}$$

Bounding the objective value

$\pi_{\theta'}$ is *close* to π_{θ} if $\frac{1}{2} \sum_{\mathbf{a}_t} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)| \leq \epsilon$ for all $\mathbf{s}_t$

$$\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_{\theta}(\mathbf{s}_t)| \leq \epsilon t$$

$$\begin{aligned} E_{p_{\theta'}(\mathbf{s}_t)}[f(\mathbf{s}_t)] &= \sum_{\mathbf{s}_t} p_{\theta'}(\mathbf{s}_t) f(\mathbf{s}_t) \geq \sum_{\mathbf{s}_t} p_{\theta}(\mathbf{s}_t) f(\mathbf{s}_t) - |p_{\theta}(\mathbf{s}_t) - p_{\theta'}(\mathbf{s}_t)| \max_{\mathbf{s}_t} f(\mathbf{s}_t) \\ &\geq E_{p_{\theta}(\mathbf{s}_t)}[f(\mathbf{s}_t)] - 2\epsilon t \max_{\mathbf{s}_t} f(\mathbf{s}_t) \end{aligned}$$

$$\begin{aligned} \sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] &\geq \\ \sum_t E_{\mathbf{s}_t \sim p_{\theta}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] &- \sum_t 2\epsilon t C \end{aligned}$$

$O(Hr_{\max})$ or $O\left(\frac{r_{\max}}{1-\gamma}\right)$



maximizing this maximizes a bound on the thing we want!

Where are we at so far?

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

such that $D_{\text{TV}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t), \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$

for small enough ϵ , this is guaranteed to improve $J(\theta') - J(\theta)$

This means it's “OK” to ignore the state distribution
if we do not change the policy “too much”

Intermission

Part 4:

Policy gradient with constraints

A more convenient bound

Claim: $p_\theta(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_θ is *close* to $\pi_{\theta'}$

$\pi_{\theta'}$ is *close* to π_θ if $\frac{1}{2} \sum_{\mathbf{s}} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)| \leq \epsilon$ for all $\mathbf{s}_t$

$$\frac{1}{2} \sum_{\mathbf{s}} |p_{\theta'}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| \leq 2\epsilon t$$

a more convenient bound: $\frac{1}{2} \sum_{\mathbf{s}_t} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)| \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t))}$

$\Rightarrow D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t))$ bounds state marginal difference

$$D_{\text{KL}}(p_1(x) \parallel p_2(x)) = E_{x \sim p_1(x)} \left[\log \frac{p_1(x)}{p_2(x)} \right]$$

KL divergence has some very convenient properties that make it much easier to approximate!

Aside: KL divergence

$D_{\text{KL}}(\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t))$ bounds state marginal difference

$$D_{\text{KL}}(p_1(x) \parallel p_2(x)) = E_{x \sim p_1(x)} \left[\log \frac{p_1(x)}{p_2(x)} \right] \approx \sum_i \log p_1(x_i) - \log p_2(x_i)$$

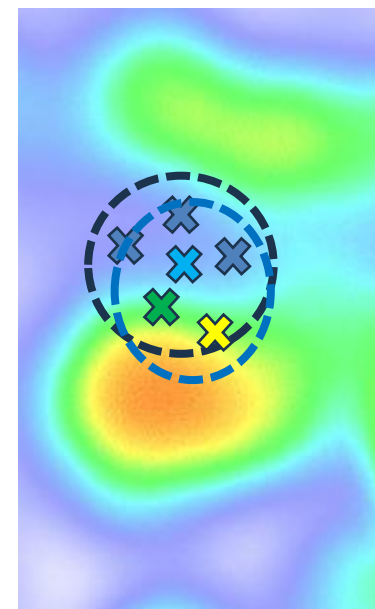
using $x_i \sim p_1(x)$

$$D_{\text{KL}}(\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) \approx \sum_i \log \pi_{\theta}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)}) - \log \pi_{\theta'}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})$$

using $(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) \sim \pi_{\theta}(\mathbf{s}_t, \mathbf{a}_t)$

doesn't depend on θ'
(the optimization variable)

minimizing $D_{\text{KL}}(\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t))$ w.r.t. θ' simply means
maximizing the likelihood of samples from $\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)$



Where are we at so far?

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

such that $D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$

for small enough ϵ , this is guaranteed to improve $J(\theta') - J(\theta)$

How do we enforce the constraint?

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

such that $D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$

$$\mathcal{L}(\theta', \beta) = \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] - \beta (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) - \epsilon)$$

1. Maximize $\mathcal{L}(\theta', \beta)$ with respect to θ'  **can do this incompletely (for a few grad steps)**
2. $\beta \leftarrow \beta + \alpha (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) - \epsilon)$

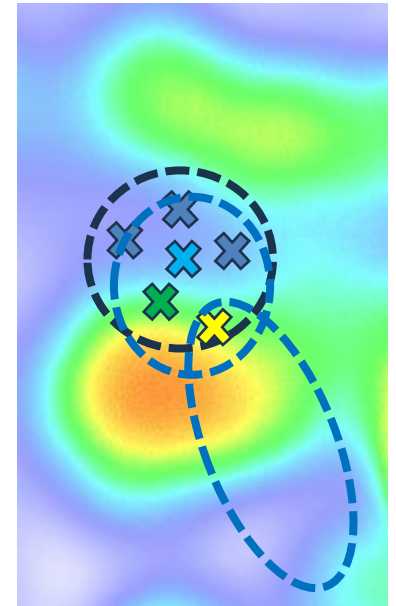
Intuition: raise β if constraint violated too much, else lower it
an instance of *dual gradient descent*

Estimating the objective with samples

$$\begin{aligned}\mathcal{L}(\theta', \beta) &= \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] - \beta (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t | \mathbf{s}_t) \| \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)) - \epsilon) \\ &\approx \underbrace{\sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} A^{\pi_\theta}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})}_{\text{importance sampled surrogate objective}} + \underbrace{\beta \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}_{\text{penalty for moving away from samples}} + \text{const}\end{aligned}$$

importance sampled surrogate objective

penalty for moving
away from samples

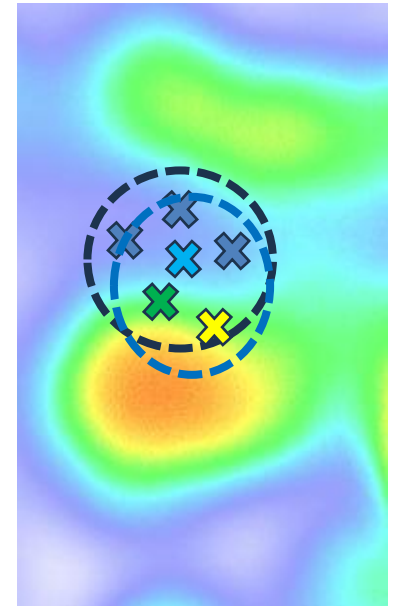


PPO with KL divergence

policy gradient with KL-constraint (PPO):

1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
5. $\theta' \leftarrow \theta$
- $K \times$
 6. $\nabla_{\theta'} J(\theta') \approx \nabla_{\theta'} \mathcal{L}_{\text{KL}}(\theta')$
 7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
6. $\beta \leftarrow \beta + \alpha (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) - \epsilon)$
7. $\theta \leftarrow \theta'$

$$\mathcal{L}_{\text{KL}}(\theta', \beta) = \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})} A^{\pi_\theta}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \beta \log \pi_{\theta'}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})$$



Part 4:

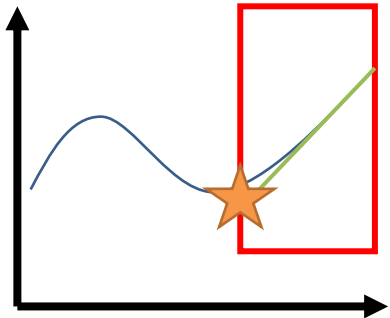
Natural policy gradient

How (else) do we optimize the objective?

$$\theta' \leftarrow \arg \max_{\theta'} \overbrace{\sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]}^{\bar{A}(\theta')}$$

such that $D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$

for small enough ϵ , this is guaranteed to improve $J(\theta') - J(\theta)$



$$\theta' \leftarrow \arg \max_{\theta'} \nabla_\theta \bar{A}(\theta)^T (\theta' - \theta)$$

such that $D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$

Use first order Taylor approximation for objective (a.k.a., linearization)

How do we optimize the objective?

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

such that $D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$

$$\theta' \leftarrow \arg \max_{\theta'} \nabla_\theta \bar{A}(\theta)^T (\theta' - \theta)$$

such that $D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$

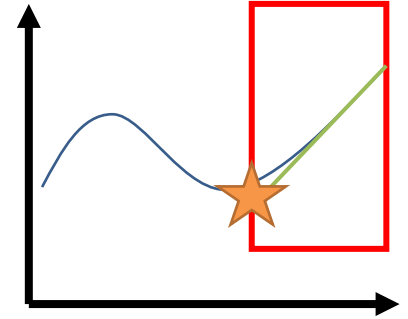
$$\nabla_{\theta'} \bar{A}(\theta') = \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

(see policy gradient lecture for derivation)

$$\nabla_\theta \bar{A}(\theta) = \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\cancel{\frac{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t | \mathbf{s}_t)}} \nabla_\theta \log \pi_\theta(\mathbf{a}_t | \mathbf{s}_t) A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

$$\nabla_\theta \bar{A}(\theta) = \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} \left[\nabla_\theta \log \pi_\theta(\mathbf{a}_t | \mathbf{s}_t) A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] = \nabla_\theta J(\theta)$$

exactly the normal policy gradient!



Can we just use the gradient then?

$$\theta' \leftarrow \arg \max_{\theta'} \nabla_{\theta} J(\theta)^T (\theta' - \theta)$$

$$\text{such that } D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$$

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$$

$$\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$$

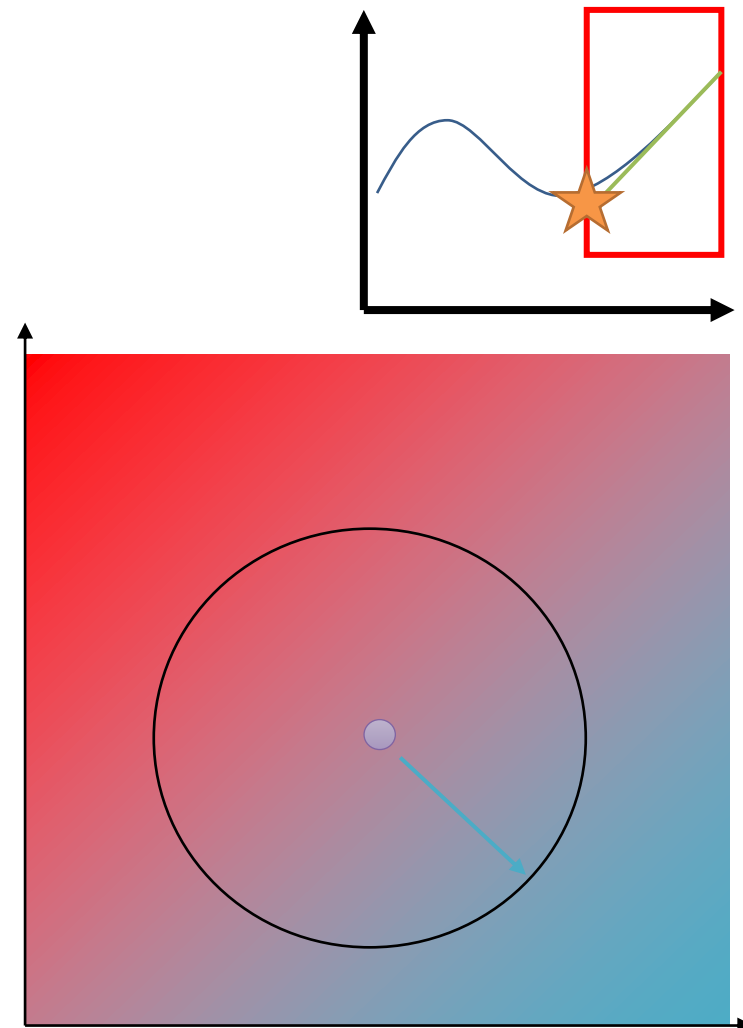
some parameters change probabilities a lot more than others!

Claim: gradient ascent does this:

$$\theta' \leftarrow \arg \max_{\theta'} \nabla_{\theta} J(\theta)^T (\theta' - \theta)$$

$$\text{such that } \|\theta - \theta'\|^2 \leq \epsilon$$

$$\theta' = \theta + \sqrt{\frac{\epsilon}{\|\nabla_{\theta} J(\theta)\|^2}} \nabla_{\theta} J(\theta)$$



Can we just use the gradient then?

$$\theta' \leftarrow \arg \max_{\theta'} \nabla_{\theta} J(\theta)^T (\theta' - \theta)$$

such that $D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$



not the same!

$$\theta' \leftarrow \arg \max_{\theta'} \nabla_{\theta} J(\theta)^T (\theta' - \theta)$$

such that $\|\theta - \theta'\|^2 \leq \epsilon$

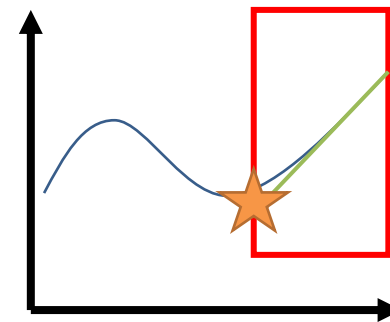
second order Taylor expansion

$$D_{\text{KL}}(\pi_{\theta'} \parallel \pi_{\theta}) \approx \frac{1}{2} (\theta' - \theta)^T \mathbf{F} (\theta' - \theta)$$

Fisher-information matrix

$$\mathbf{F} = E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(\mathbf{a}|\mathbf{s}) \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}|\mathbf{s})^T]$$

can estimate with samples



Can we just use the gradient then?

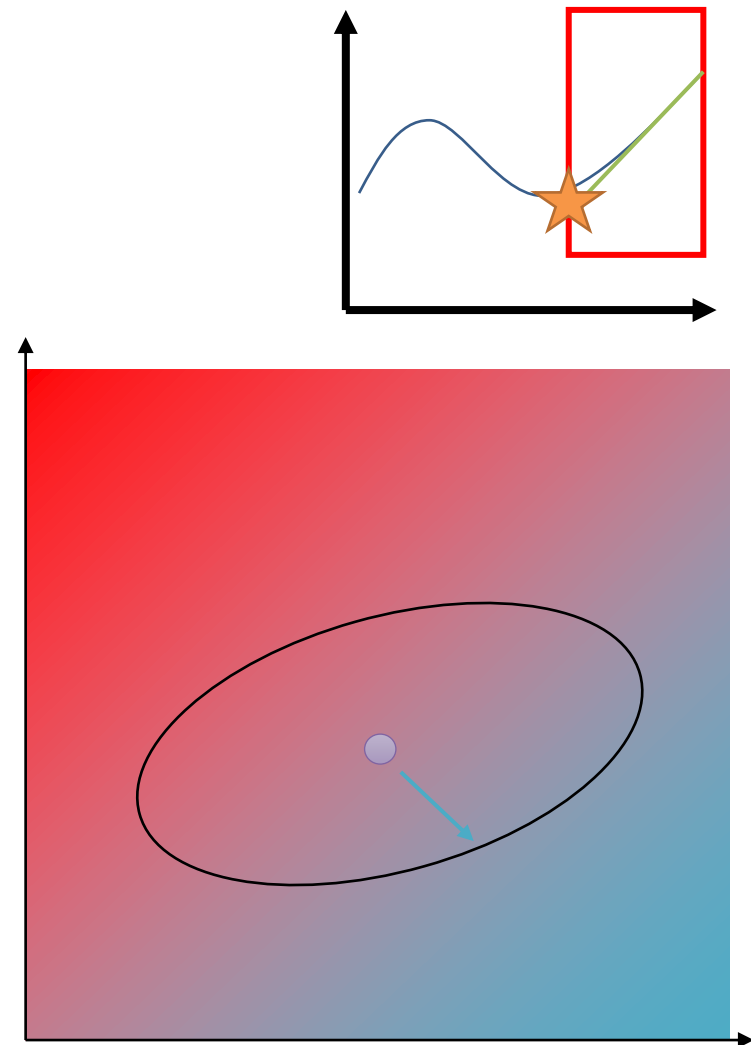
$$\theta' \leftarrow \arg \max_{\theta'} \nabla_{\theta} J(\theta)^T (\theta' - \theta)$$

$$\text{such that } D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \| \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)) \leq \epsilon$$

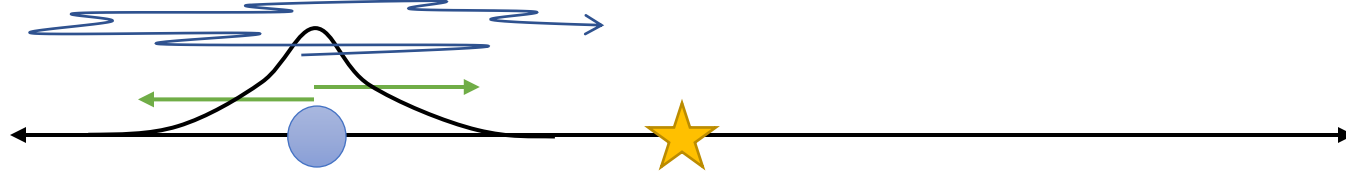
$$D_{\text{KL}}(\pi_{\theta'} \| \pi_{\theta}) \approx \frac{1}{2} (\theta' - \theta)^T \mathbf{F} (\theta' - \theta)$$

$$\theta' = \theta + \alpha \mathbf{F}^{-1} \nabla_{\theta} J(\theta) \quad \leftarrow \text{natural gradient}$$

$$\alpha = \sqrt{\frac{2\epsilon}{\nabla_{\theta} J(\theta)^T \mathbf{F} \nabla_{\theta} J(\theta)}}$$



Is this even a problem in practice?

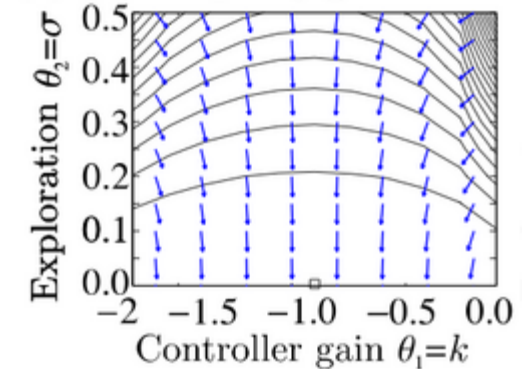


$$r(\mathbf{s}_t, \mathbf{a}_t) = -\mathbf{s}_t^2 - \mathbf{a}_t^2$$

$$\log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) = -\frac{1}{2\sigma^2} (k\mathbf{s}_t - \mathbf{a}_t)^2 + \text{const}$$

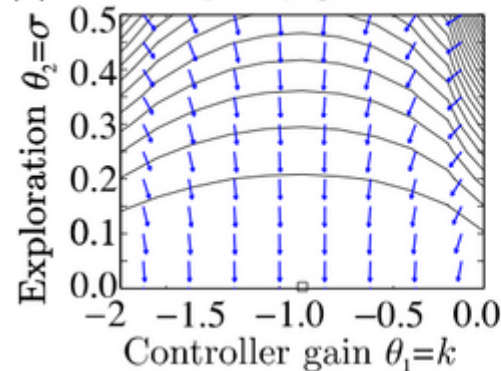
$$\theta = (k, \sigma)$$

(a) 'Vanilla' policy gradients

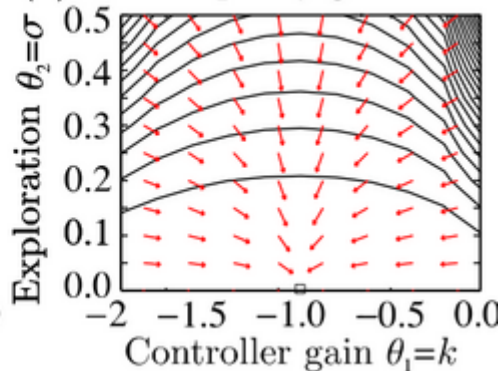


(image from Peters & Schaal 2008)

(a) 'Vanilla' policy gradients

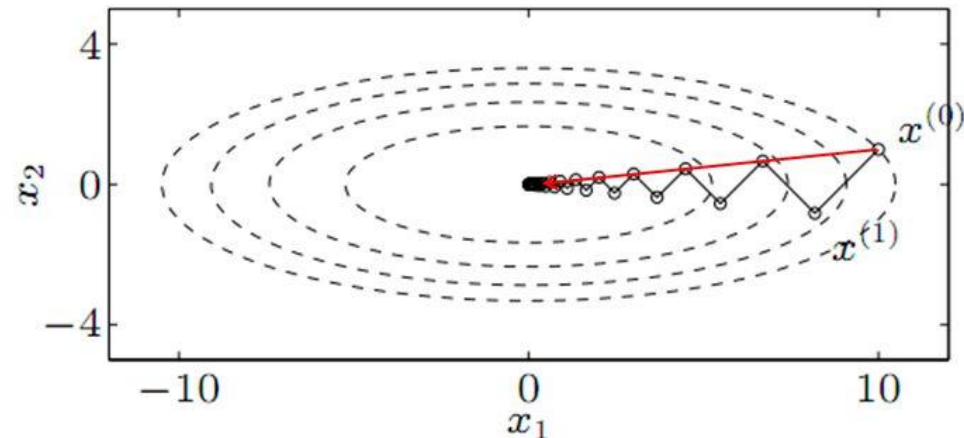


(b) Natural policy gradients



(figure from Peters & Schaal 2008)

Essentially the same problem as this:



TRPO

policy gradient with natural gradient (TRPO):

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\theta \leftarrow \theta + \alpha \mathbf{F}^{-1} \nabla_\theta J(\theta)$

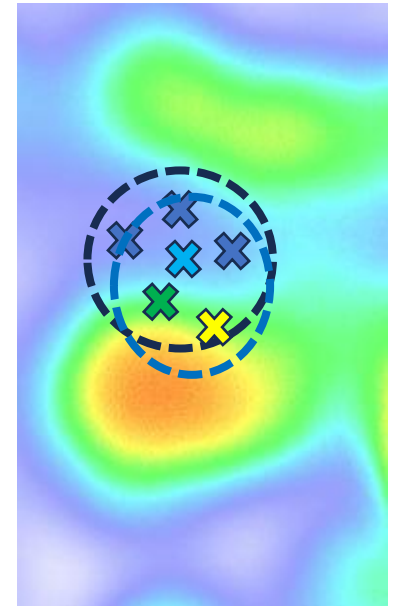
In practice, don't build the full matrix $\mathbf{F}$

Use conjugate gradient (CG) to solve the linear system in $O(n)$ time

$$\alpha = \sqrt{\frac{2\epsilon}{\nabla_\theta J(\theta)^T \mathbf{F} \nabla_\theta J(\theta)}}$$

$$\mathbf{F} \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \nabla_\theta \log \pi_\theta(\mathbf{a}_t^i | \mathbf{s}_t^i) \nabla_\theta \log \pi_\theta(\mathbf{a}_t^i | \mathbf{s}_t^i)^T$$

$$\nabla_\theta J(\theta) \approx \sum_i \sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$$



Part 5:

Review and practical guidelines

What did we learn?

- Policy gradient is *more productively* interpreted as a kind of policy iteration
 - This gives us a better explanation for why it's “OK” to optimize the policy a lot even using the same set of samples
- PPO with clipping is a heuristic approximation to PPO with KL
- There are other ways to enforce the constraint
 - Natural gradient
- TRPO vs PPO?
 - One big step vs many little ones?



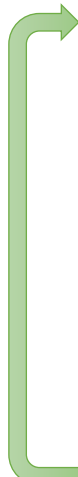
Summary

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

$$\text{such that } D_{\text{KL}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$$

for small enough ϵ , this is guaranteed to improve $J(\theta') - J(\theta)$

policy gradient with KL-constraint (PPO):

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t(i)) = \sum_{t'=t}^\infty (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\theta' \leftarrow \theta$
 -  6. $\nabla_{\theta'} J(\theta') \approx \nabla_{\theta'} \mathcal{L}_{\text{KL}}(\theta')$
 -  7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
 6. $\beta \leftarrow \beta + \alpha (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) - \epsilon)$
 7. $\theta \leftarrow \theta'$

policy gradient with natural gradient (TRPO):

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t(i)) = \sum_{t'=t}^\infty (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\theta \leftarrow \theta + \alpha \mathbf{F}^{-1} \nabla_\theta J(\theta)$